

LECTURE 5: MATHEMATICAL EXPECTATION AND RELATED BOUNDS

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to `elre@princeton.edu`, thanks in advance!

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Definition 5.1. Let X be a random variable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The **expectation** (or expected value) of X is defined as the Lebesgue integral $\mathbb{E}[X] := \int_{\Omega} X \, d\mathbb{P}$ (when it exists). We say that a random variable X is (absolutely) integrable if $\int |X| \, d\mathbb{P} < \infty$; that is, if it is integrable as a measurable function.

Some properties of expectation are immediately inherited from the properties of Lebesgue integral, such as

- Expectations of constants: $\mathbb{E}[c] = c\mathbb{E}[\mathbf{1}_{\Omega}] = c\mathbb{P}(\Omega) = c$.
- Linearity: $\mathbb{E}[\alpha X + \beta Y] = \alpha\mathbb{E}[X] + \beta\mathbb{E}[Y]$ almost surely.
- Monotonicity: if $X \leq Y$ a.s., then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.

Recall that products (and powers) of random variables are also random variables.

Definition 5.2 (Moments). If k is a positive integer, then the **k -th moment** of X is defined by $\mathbb{E}X^k$ (when it exists). If $\mathbb{E}X^2 < \infty$, the **variance** of X is defined by $\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X]^2 - (\mathbb{E}X)^2$. For $1 \leq q < \infty$, we denote by $L^q(\Omega, \mathcal{F}, \mathbb{P})$ the collection of random variables X on $(\Omega, \mathcal{F}, \mathbb{P})$ for which $\|X\|_{L^q} := [\mathbb{E}|X|^q]^{1/q} < \infty$.

Two more straightforward observations:

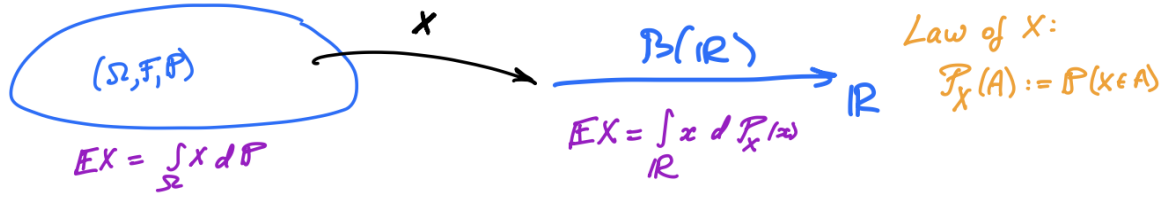
- A random variable belongs to $L^k(\Omega, \mathcal{F}, \mathbb{P})$ if and only if it has (finite) k -th moment.
- If $X \in L^2(\Omega, \mathcal{F}, \mathbb{P})$ and $\alpha, \beta \in \mathbb{R}$, then $\text{Var}(\alpha X + \beta) = \alpha^2 \text{Var}(X)$.

This explains how to compute the expectations of constants, indicators and simple functions, e.g.,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A) \quad \text{and} \quad \mathbb{E}\left[\sum_{i=1}^N c_i \mathbf{1}_{A_i}\right] = \sum_{i=1}^N c_i \mathbb{P}(A_i).$$

However, it is not immediately clear how to compute the expectation of a general real-valued random variable, which implicitly requires taking the supremum over all lower-bounding non-negative simple functions (what would be the value in each case?).

For example, how would we compute the expectation of a distribution that only takes finitely or countably many values, e.g., a uniform distribution with $\mathbb{P}(X = k) = 1/T$ for $k = 1, \dots, T$, or a geometric distribution with success probability $p \in (0, 1)$ and $\mathbb{P}(X = k) = p(1 - p)^{k-1}$? *What are the sets A_i ? What is the space Ω ? Do we even really care?*



Theorem 5.3 (Change of variables). Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a random variable with law $\mathcal{P}_X$ and $f : \mathbb{R} \rightarrow \bar{\mathbb{R}}$ be a measurable function such that $f(X)$ is integrable. Then

$$\int_{\Omega} f(X(\omega)) d\mathbb{P}(\omega) = \int_{\mathbb{R}} f(x) d\mathcal{P}_X(x).$$

Proof. The proof goes via the standard 4-step argument.

- (1) If $f = \mathbb{1}_A$ is an indicator function, then

$$\int_{\Omega} \mathbb{1}_A(X(\omega)) d\mathbb{P} = \mathbb{P}(\{\omega : X(\omega) \in A\}) = \mathcal{P}_X(A) = \int_{\mathbb{R}} \mathbb{1}_A(x) d\mathcal{P}_X.$$

- (2) If $f \in \mathcal{SF}_+(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ is a non-negative simple function, then we use the linearity of the integrals above.
- (3) To extend to non-negative measurable functions, we use the monotone convergence theorem: if $f_n \rightarrow f$ increasing pointwise, then we also have $f_n(X(\omega)) \rightarrow f(X(\omega))$ increasing pointwise and we can take this limit outside of the integral.
- (4) Finally, to extend to integrable functions, we write $f = f_+ - f_-$ and use linearity again. $\square$

In particular, if two random variables X and Y have the same law, then $\mathbb{E}[X]$ exists if and only if $\mathbb{E}[Y]$ exists, and in this case they are equal. *So, the expectation is a property of the law of a random variable, rather than of the random variable itself* (in other words, we do not really need to care about the underlying probability space Ω at all).

Using Theorem 5.3, we obtain the following properties in particular:

- Calculating moments:

$$\mathbb{E}[X^q] = \int_{\mathbb{R}} x^q d\mathcal{P}_X(x).$$

- If a random variable has a probability density function f_X , then $\mathcal{P}_X(A) = \int_A f_X(x) dx$ and $d\mathcal{P}_X(x) = f_X dx$, and so

$$\mathbb{E}[X] = \int_{\mathbb{R}} x d\mathcal{P}_X(x) = \int_{\mathbb{R}} x f_X(x) dx.$$

(See, e.g., [Dembo, Corollary 1.3.62] for the full technical details.) This gives us the usual way of computing the expectations and higher moments of distributions with densities such as the Gaussian distribution, exponential distribution, etc.

AN EXAMPLE COMPUTATION FOR A DISCRETE DISTRIBUTION

Let us consider a random variable X with the geometric distribution $\text{Geo}(p)$ with success probability $p \in (0, 1)$, such that $\mathbb{P}(X = k) = p(1 - p)^{k-1}$ for $k = 1, 2, \dots$. Recall that the geometric distribution models the number of independent coin tosses needed until the first

appearance of a head (or more generally, the number of independent trials until the first occurrence in this sequence of a specific event whose probability is p).

The laws of discrete distributions can be described with their values at individual points. In particular,

$$\mathcal{P}_X(A) \stackrel{(1)}{=} \sum_{k \in A} \mathcal{P}_X(\{k\}) = \sum_{k=1}^{\infty} \mathbb{1}_{\{k \in A\}} \mathbb{P}(X = k) = \sum_{k=1}^{\infty} p(1-p)^{k-1} \mathbb{1}_{\{k \in A\}}.$$

Due to the equality (1), the laws of a discrete random variable are often characterized by their *probability mass functions* $f(x) := \mathcal{P}_X(\{x\})$. The computation above shows that for $X \sim \text{Geo}(p)$, we have

$$f(x) = \sum_{k=1}^{\infty} p(1-p)^{k-1} \mathbb{1}_{\{x=k\}}.$$

(Note that only one indicator can be nonzero for each x .)

Now, let's compute the expectation and check that it agrees with the usual formula for discrete distributions:

$$\mathbb{E}[X] = \int_{\mathbb{R}} x \, d\mathcal{P}_X(x) = \sum_{k=1}^{\infty} kp(1-p)^{k-1}.$$

This can be formally checked as follows. Consider any non-negative simple function $g(x)$ that lower bounds x ; i.e., of the form $g(x) = \sum_{i=1}^N c_i \mathbb{1}_{A_i}$ such that $g(x) \leq x$. Then

$$\begin{aligned} \int_{\mathbb{R}} x \, d\mathcal{P}_X(x) &= \sup_{g(x) \leq x} \left\{ \sum_{i=1}^N c_i \mathcal{P}_X(A_i) \right\} \\ &= \sup_{g(x) \leq x} \left\{ \sum_{i=1}^N c_i \sum_{k=1}^{\infty} \mathcal{P}_X(\{k\}) \mathbb{1}_{\{k \in A_i\}} \right\} \\ &= \sup_{g(x) \leq x} \left\{ \sum_{k=1}^{\infty} \mathcal{P}_X(\{k\}) \sum_{i=1}^N c_i \mathbb{1}_{\{k \in A_i\}} \right\} \\ &= \sum_{k=1}^{\infty} \mathcal{P}_X(\{k\}) k \end{aligned}$$

since we must have $c_i \leq k$ if $k \in A_i$ (as $g(x) = \sum_{i=1}^N c_i \mathbb{1}_{A_i}(x) \leq x$). Taking supremum ensures that we can cover all positive integers k by some A_i (in the limit) and can take $c_i = k$.

A similar check shows that

$$\mathbb{E}[X(X-1)] = \int_{\mathbb{R}} x(x-1) \, d\mathcal{P}_X(x) = \sum_{k=1}^{\infty} k(k-1)p(1-p)^{k-1}.$$

From here, verify that $\mathbb{E}[X] = p^{-1}$ and $\text{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - (\mathbb{E}[X])^2 = (1-p)/p^2$. Intuitively, the smaller the probability p of heads is, the longer we expect to wait until the first appearance of a head, and thus the variance of the first appearance is also larger.

INEQUALITIES BOUNDING EXPECTATIONS

Recall that a function $g : \mathbb{R} \rightarrow \mathbb{R}$ is **convex** if for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$, $g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y)$.

Theorem 5.4 (Jensen's inequality). *Let $g : \mathcal{I} \rightarrow \mathbb{R}$ be a measurable convex function on an open interval $\mathcal{I} \subseteq \mathbb{R}$, and X be a random variable such that $X \in \mathcal{I}$ almost surely and $\mathbb{E}[g(X)]$ exists. Then*

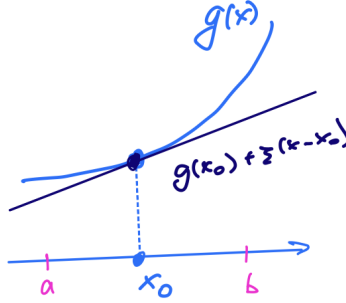
$$\mathbb{E}[g(X)] \geq g(\mathbb{E}[X]).$$

Proof. The proof follows from two generic properties of convex functions:

- Claim 1: A convex function on an open set must be continuous (exercise!), so $g(X)$ is measurable.
- Claim 2: For any $x_0 \in \mathcal{I}$, there exists $\xi \in \mathbb{R}$ such that

$$g(x) \geq g(x_0) + \xi(x - x_0) \quad \text{for any } x \in \mathcal{I}.$$

This essentially says that a convex function can always be lower bounded by a line (e.g., if g is differentiable, it is lower bounded by its tangent lines).



- From Claim 2, we can substitute x with the random variable $X(\omega)$ and take the expectation. By linearity of expectation, we have:

$$\mathbb{E}g(X) \geq g(x_0) + \xi(\mathbb{E}X - x_0).$$

Take $x_0 = \mathbb{E}X$ to conclude that $\mathbb{E}g(X) \geq g(\mathbb{E}X) + 0$.

The proof of Claim 2 is as follows: recall that by definition of a convex function, for any $x, y \in \mathcal{I}$ and any $\lambda \in (0, 1)$, we have

$$g(\lambda x + (1 - \lambda)y) \leq \lambda g(x) + (1 - \lambda)g(y).$$

Fix $x_0 \in \mathcal{I}$. Applying the definition to $x = x_0 - \varepsilon_-$ and $y = x_0 + \varepsilon_+$ for $\varepsilon_-, \varepsilon_+ > 0$ (small enough such that $x, y \in \mathcal{I}$), and $\lambda = \varepsilon_+ / (\varepsilon_- + \varepsilon_+)$, we get

$$\frac{\varepsilon_+}{\varepsilon_- + \varepsilon_+} g(x_0 - \varepsilon_-) + \frac{\varepsilon_-}{\varepsilon_- + \varepsilon_+} g(x_0 + \varepsilon_+) \geq g(x_0).$$

Thus, by rearranging, we deduce that

$$\frac{g(x_0 + \varepsilon_+) - g(x_0)}{\varepsilon_+} \geq \frac{g(x_0) - g(x_0 - \varepsilon_-)}{\varepsilon_-}.$$

Choose any $\xi \in \mathbb{R}$ such that

$$\xi \in \left[\sup_{\varepsilon_- > 0} \frac{g(x_0) - g(x_0 - \varepsilon_-)}{\varepsilon_-}, \inf_{\varepsilon_+ > 0} \frac{g(x_0 + \varepsilon_+) - g(x_0)}{\varepsilon_+} \right].$$

(Note that this interval is non-empty by the preceding displayed equation.) Thus, for any $\varepsilon > 0$, we have

$$\frac{g(x_0) - g(x_0 - \varepsilon)}{\varepsilon} \leq \xi \leq \frac{g(x_0 + \varepsilon) - g(x_0)}{\varepsilon},$$

from which we conclude that $g(x) \geq g(x_0) + \xi(x - x_0)$ for both cases $x < x_0$ and $x > x_0$, which completes the proof of Claim 2. $\square$

Corollary 5.5 (Nesting of L^p spaces). *Suppose that $1 \leq p \leq q \leq \infty$. Then $L^q(\Omega, \mathcal{F}, \mathbb{P}) \subseteq L^p(\Omega, \mathcal{F}, \mathbb{P})$. Moreover, $\|X\|_{L^p} \leq \|X\|_{L^q}$.*

Proof. Suppose that $X \in L^q(\Omega, \mathcal{F}, \mathbb{P})$, i.e., $\|X\|_{L^q} = (\mathbb{E}|X|^q)^{1/q} < \infty$. Note that since $q/p \geq 1$, the function $g(x) = |x|^{q/p}$ is convex on $\mathbb{R}$. Thus, by Jensen's inequality (Theorem 5.4) applied to the random variable X^p and convex function g ,

$$(\mathbb{E}|X|^p)^{q/p} = g(\mathbb{E}[X^p]) \leq \mathbb{E}[g(X^p)] = \mathbb{E}|X|^q < \infty.$$

Exponentiating by $1/q$ on both sides shows that $\|X\|_{L^p} \leq \|X\|_{L^q} < \infty$, so $X \in L^p(\Omega, \mathcal{F}, \mathbb{P})$. $\square$

Theorem 5.6 (Hölder's inequality). *For any $p, q \geq 1$ such that $\frac{1}{p} + \frac{1}{q} = 1$ we have*

$$\mathbb{E}|XY| \leq \|X\|_{L^p} \cdot \|Y\|_{L^q}.$$

One way to prove Hölder's inequality is to use Young's inequality:

Lemma 5.7 (Young's inequality). *If $a \geq 0, b \geq 0$ and $p, q \geq 1$ are such that $\frac{1}{p} + \frac{1}{q} = 1$, then*

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Proof. We may assume that $a, b > 0$, otherwise the bound is trivially true. Let $\lambda = 1/p$, so $1 - \lambda = 1/q$. Since $x \mapsto \log(x)$ is concave (so $x \mapsto -\log(x)$ is convex),

$$\log(\lambda a^p + (1 - \lambda)b^q) \geq \lambda \log(a^p) + (1 - \lambda) \log(b^q) = \log(a) + \log(b) = \log(ab).$$

Exponentiating leads to the claimed bound. $\square$

Proof of Theorem 5.6. By homogeneity, we may assume that $\|X\|_{L^p} = \|Y\|_{L^q} = 1$. Thus, we want to prove that $\mathbb{E}|XY| \leq 1$. By Young's inequality (Lemma 5.7),

$$|XY| \leq \frac{|X|^p}{p} + \frac{|Y|^q}{q}.$$

Taking the expectation on both sides shows that

$$\mathbb{E}|XY| \leq \frac{\mathbb{E}|X|^p}{p} + \frac{\mathbb{E}|Y|^q}{q} = \frac{1}{p} + \frac{1}{q} = 1. \quad \square$$

Remark 5.8. • If the goal is to estimate the expectation of the product of random variables (and not the absolute value of the product), note that as an easy corollary of Jensen's inequality (how?), we have

$$|\mathbb{E}[XY]| \leq \mathbb{E}|XY| \leq \|X\|_{L^p} \cdot \|Y\|_{L^q}. \quad (1)$$

- Note that by choosing $p = q = 2$ in particular, we obtain

$$\mathbb{E}|XY| \leq \sqrt{\mathbb{E}[X^2]} \cdot \sqrt{\mathbb{E}[Y^2]}. \quad (2)$$

This is the important **Cauchy-Schwarz inequality**. Note that in other applications of Hölder's inequality, one of the pair p and q is less than 2 and the other is larger than 2.

Theorem 5.9 (Minkowski's inequality). *For any $p \geq 1$, if $X, Y \in L^p(\Omega, \mathcal{F}, \mathbb{P})$, then*

$$\|X + Y\|_{L^p} \leq \|X\|_{L^p} + \|Y\|_{L^p}.$$

Proof. First, note that $X + Y \in L^p(\Omega, \mathcal{F}, \mathbb{P})$ from the inequality $|X + Y|^p \leq 2^{p-1}(|X|^p + |Y|^p)$ (which follows from the convexity of $x \mapsto |x|^p$ for $p \geq 1$). Moreover, we may assume that $\|X + Y\|_{L^p} > 0$ (otherwise, the inequality holds trivially).

Note that by the triangle inequality $|X + Y| \leq |X| + |Y|$,

$$\|X + Y\|_{L^p}^p \leq \| |X| \cdot |X + Y|^{p-1} \|_{L^1} + \| |Y| \cdot |X + Y|^{p-1} \|_{L^1}.$$

By Hölder's inequality (Theorem 5.6) with $q = \frac{p}{p-1}$,

$$\| |X| \cdot |X + Y|^{p-1} \|_{L^1} \leq \|X\|_{L^p} \cdot \|X + Y\|_{L^p}^{p-1},$$

and similarly, $\| |Y| \cdot |X + Y|^{p-1} \|_{L^1} \leq \|Y\|_{L^p} \cdot \|X + Y\|_{L^p}^{p-1}$. By combining these bounds and rearranging, we obtain $\|X + Y\|_{L^p}^p \leq \|X\|_{L^p}^p + \|Y\|_{L^p}^p$. $\square$

Observe that $\|X\|_{L^p} = (\mathbb{E}|X|^p)^{1/p}$ defines a seminorm on the L^p -space $L^p(\Omega, \mathcal{F}, \mathbb{P})$. This means that it is a non-negative symmetric function that satisfies the triangle inequality, i.e., Minkowski's inequality (Theorem 5.9), but it can be zero on a nonzero X . (However, note that it is precisely zero on the set of random variables that are almost surely zero.) Moreover, Minkowski's inequality can also be used to bound expectations of some polynomial functions of a random variable.

CONCENTRATION INEQUALITIES: EXPECTATIONS AND MOMENTS HELP QUANTIFY THE BEHAVIOUR OF A RANDOM VARIABLE

In the following, we will not aim to bound the expectation of a random variable X , but use the expectation (or higher moments) of X to estimate its law. In particular, it is important to bound how large X can be, i.e., estimate *the tails* of its distribution. The following theorem gives a very handy tool to do so as we will see from the special cases below:

Theorem 5.10 (General Markov's inequality). *If $f : \mathbb{R} \rightarrow [0, \infty]$ is a non-negative measurable function and A is a measurable set, then*

$$\left(\inf_{x \in A} f(x) \right) \cdot \mathbb{P}(X \in A) \leq \mathbb{E}[f(X) \cdot \mathbf{1}_{\{X \in A\}}] \leq \mathbb{E}[f(X)].$$

Proof. Note that for any $x \in \mathbb{R}$,

$$\left(\inf_{x \in A} f(x) \right) \cdot \mathbf{1}_{\{x \in A\}} \leq f(x) \cdot \mathbf{1}_{\{x \in A\}} \leq f(x).$$

Take the expectation of this inequality (replacing x with the random variable X) and use the monotonicity of expectation to conclude. $\square$

The general result above (Theorem 5.10) includes the following important special cases:

- **Markov's inequality.** Take $f(x) = x_+$ and $A = [a, \infty)$ for $a \geq 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}X_+}{a}. \quad (3)$$

- **Chebyshev's inequality.** Take $f(x) = x^2$ and $A = (-\infty, -a] \cup [a, \infty)$ for $a \geq 0$. Then by applying Theorem 5.10 with the random variable $X - \mathbb{E}X$, we deduce that

$$\mathbb{P}(|X - \mathbb{E}X| \geq a) \leq \frac{\text{Var}(X)}{a^2}. \quad (4)$$

Note that we can similarly consider $f(x) = |x|^q$ with $q \geq 2$ if the random variable X has more finite moments.

- **General Chernoff's inequality.** Take $f(x) = e^{\theta x}$ and $A = [a, \infty)$ for $a \geq 0$. Then

$$\mathbb{P}(X \geq a) \leq e^{-\theta a} \mathbb{E}e^{\theta X}. \quad (5)$$

This can be further optimized to select the best value of θ to minimize the upper bound. Note the exponential decay in a .

Overall, the more finite moments a random variable has, the faster the decay (with respect to a) that we can show. Note that we need $a \geq \mathbb{E}X$ for the bounds to be nontrivial.

The last theorem is a sort of “inverse Chebyshev” inequality: it provides a lower bound for the probability that a non-negative random variable is small.

Theorem 5.11 (Paley-Zygmund inequality). *Suppose that $X \geq 0$ is a non-negative random variable with finite second moment $\mathbb{E}[X^2] < \infty$.¹ Then for any $t \in [0, \mathbb{E}[X]]$,*

$$\mathbb{P}(X > t) \geq \frac{(\mathbb{E}[X] - t)^2}{\mathbb{E}[X^2]}.$$

Proof. We can write

$$\mathbb{E}[X] = \mathbb{E}[X\mathbf{1}_{\{X \leq t\}}] + \mathbb{E}[X\mathbf{1}_{\{X > t\}}].$$

We can bound the first term by $\mathbb{E}[X\mathbf{1}_{\{X \leq t\}}] \leq t$. For the second term, the Cauchy-Schwarz inequality (recall that this is a special case of Hölder's inequality (Theorem 5.6)) implies that

$$\mathbb{E}[X\mathbf{1}_{\{X > t\}}] \leq \sqrt{\mathbb{E}[X^2]} \cdot \sqrt{\mathbb{P}(X > t)}.$$

Combining these two inequalities completes the proof. $\square$

¹Note that X must have finite expectation by Corollary 5.5.